

Sesame Donuts Operating Cost Reduction Project

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Overview

Purpose – Decrease operation costs at Sesame Donuts

Intended Scale – Project spans 3 locations in Portland, Hillsboro, and Raleigh Hills

Timeframe – Data is collected over a month, appropriate data is wrangled, and conclusions are made in weeks following

Solution – Discover number of donuts that are discarded each day and to adjust the production values accordingly

Data Collection: Process

- Data was collected over 30 days
- Date, amount produced and discarded were recorded for each donut type
- Data Collection was completed by employees of Sesame Donuts
- Data was then entered into a spreadsheet and wrangled in Rstudio

SESAME DONUTS OPERATING COST PROJECT



Date: _____

Location (please circle): Raleigh Hills Hillsboro Downtown

Please list the number of donuts...

Type	Produced	Discarded
Bars		
Shells		
Rings		
Twist		
Plain		
Plain Old Fashioned		
Mini Rolls		
Glaze Donut		
Chocolate Glazed		
Mini Apple Fritter		
Old Fashioned		
Blueberry Fritter		
Blueberry Cake		
French Cruller		
Bearclaw		
Apple Fritter		
Buttermilk Bar		
Cinnamon Roll		
Devils Glazed		
Cinnamon Twist		
Sugared		
Double Chocolate		
Bavarian Cream		
Vanilla Cake		
Sesame Donut		

Data Collection: Constraints

The Largest Constraints stemmed from the data collection process:

- Small issues surrounding the legibility of collected data values
- Employee and Shop willingness to fill out form
- Slight Implementation Lag
- Short time frame
- Data does not account for Holidays, could be subject to slight skew

Overall Results

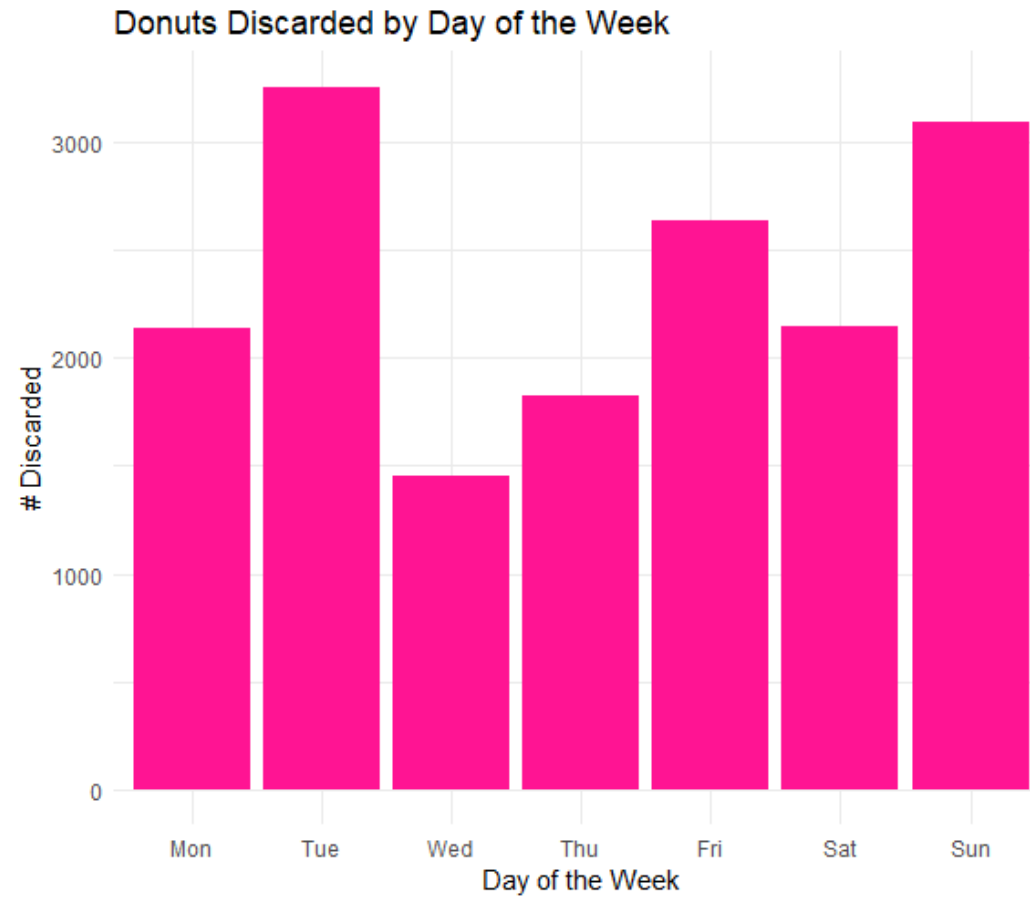
The **Total Cost** of discarded donuts over a month = \$6,867.71

The **Average Cost** of donut overproduction each day = \$236.82

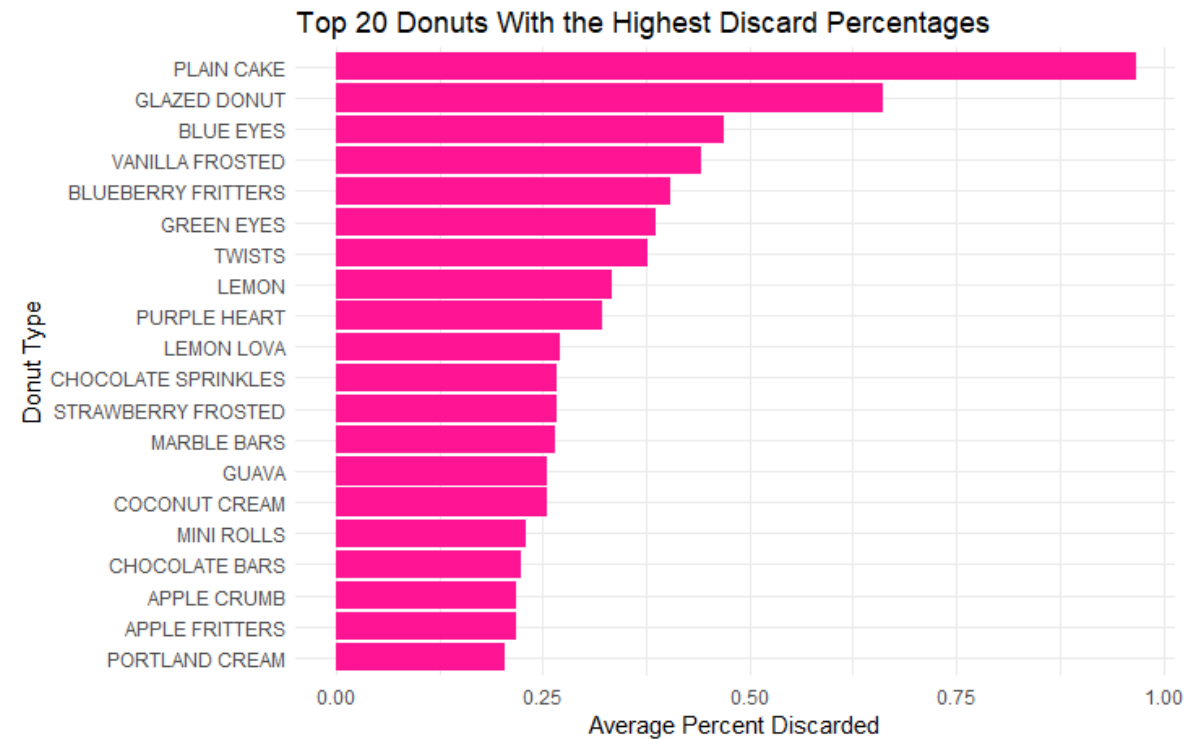
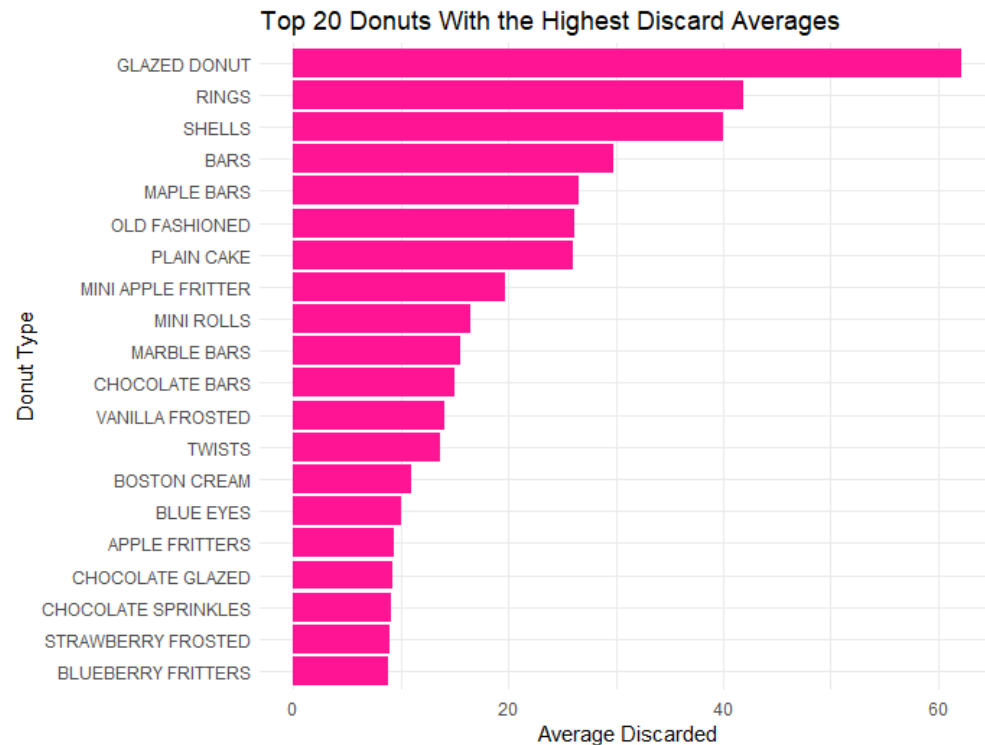
The **Most Discarded** donut = Glazed

The **Least Discarded** donut = Sid Special

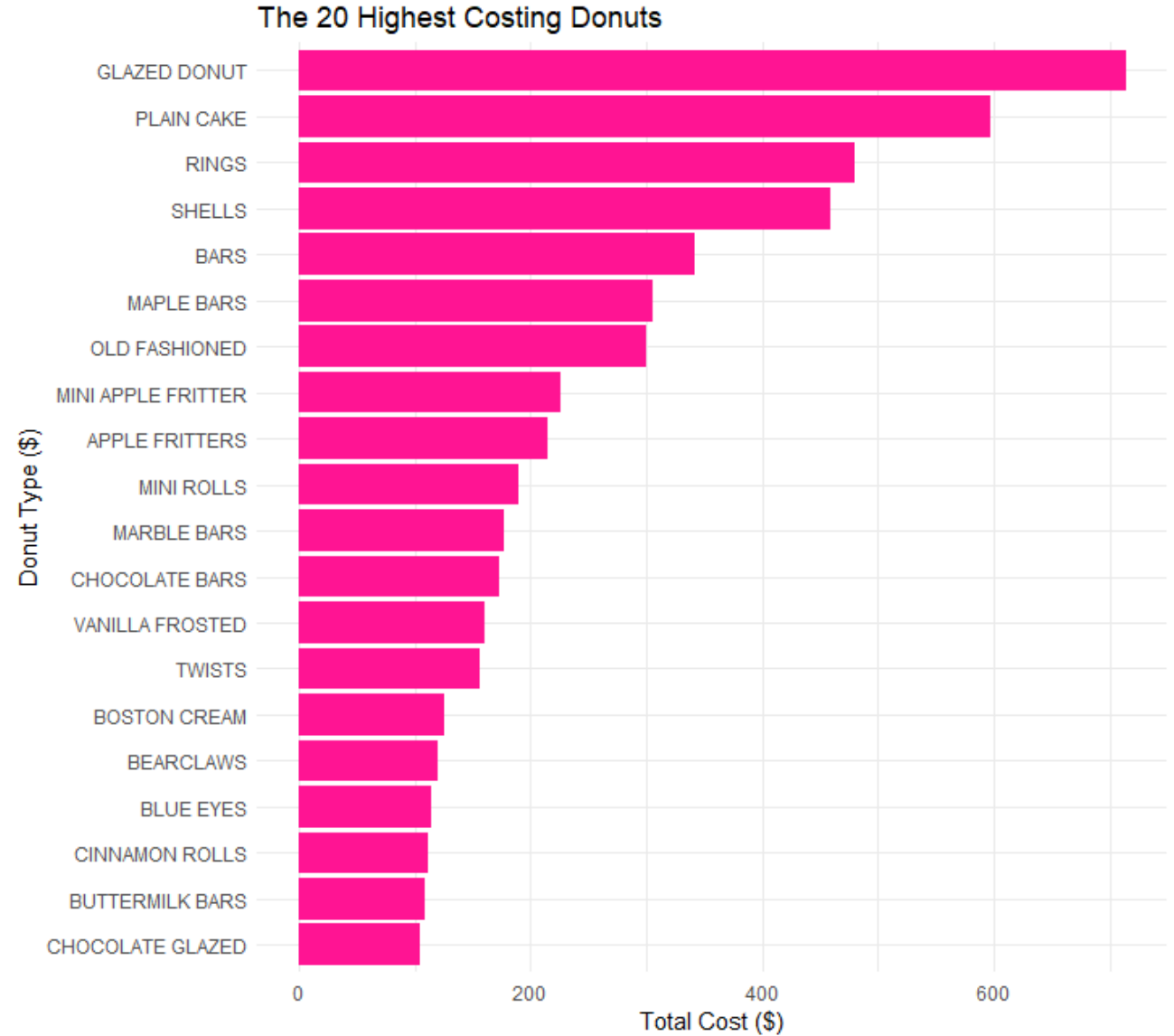
Number of Donuts Discarded Per Day of Week



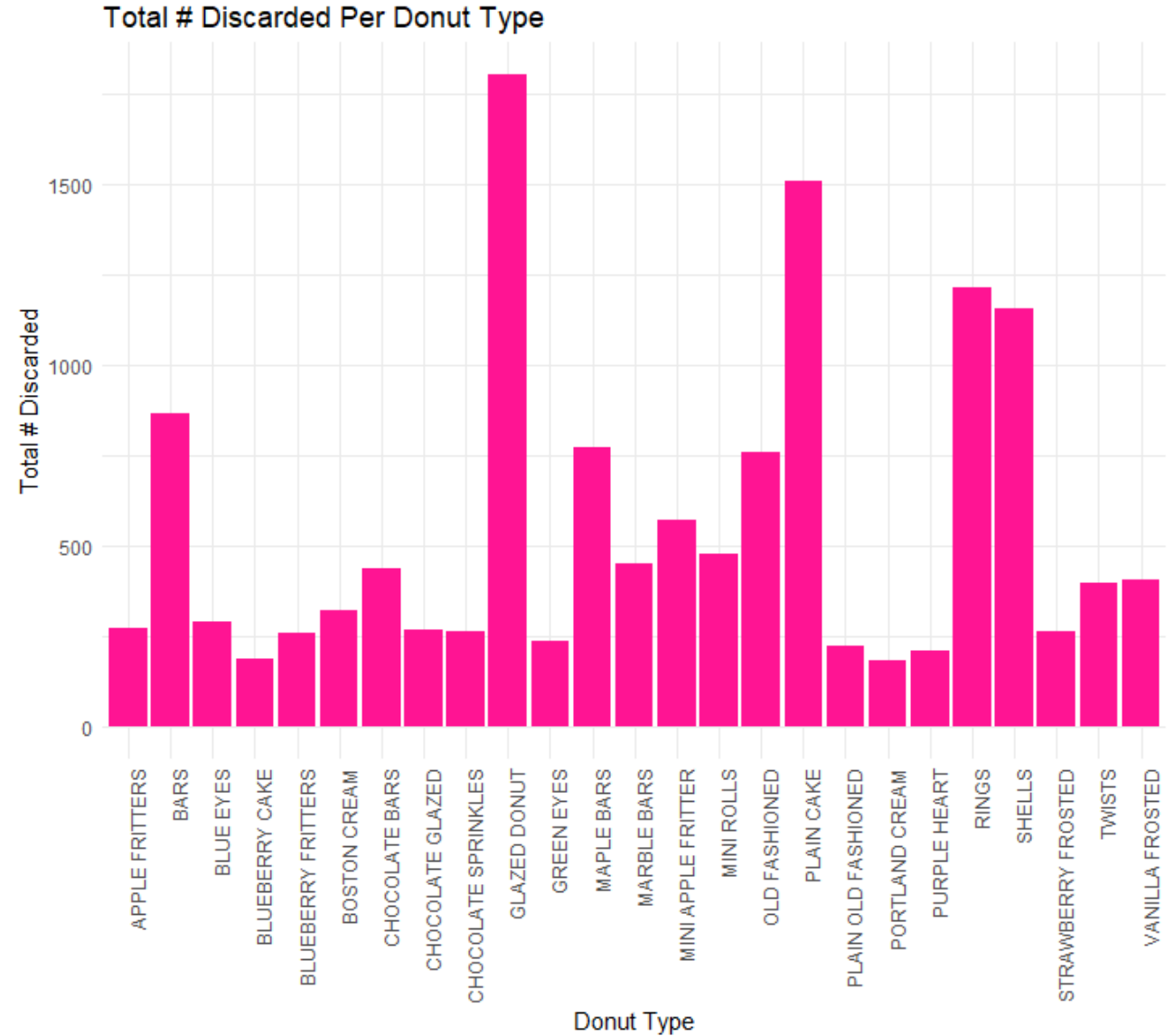
Average Number and Percentage of Different Donut Types



The 20 Highest Costing Donuts



Total Number Discarded for Most Reported Donuts



Initial Conclusions

Reduce Production by the Average Value Discarded Each Day

	DONUT_TYPE	AVG_DISCARDED
1	S'MORES	0.2068966
2	CHOCOLATE CRULLER	0.3448276
3	BARS	29.7931034
4	CHOCOLATE FROSTED	3.7586207
5	SHELLS	39.9310345
6	VANILLA CREAM	2.4482759
7	RINGS	41.8620690
8	JELLY	4.6896552
9	DEVILS GLAZED	3.4827586
10	BUTTERNUT	2.4827586
11	MINI APPLE FRITTER	19.6896552
12	MAPLE BARS	26.6206897
13	BOSTON CREAM	11.0000000
14	CINNAMON CAKE	2.6896552
15	POWDERED	3.1034483
16	SESAME DONUTS	2.9655172
17	CHOCOLATE CREAM	2.7586207
18	BAVARIAN CREAM	5.2758621
19	PORTLAND CREAM	6.2413793
20	APPLE FRITTERS	9.3448276

	DONUT_TYPE	AVG_DISCARDED
21	APPLE CRUMB	3.4827586
22	CHOCOLATE BARS	15.0344828
23	MINI ROLLS	16.5172414
24	COCONUT CREAM	3.6896552
25	GUAVA	3.4482759
26	MARBLE BARS	15.4827586
27	STRAWBERRY FROSTED	8.9655172
28	CHOCOLATE SPRINKLES	9.0344828
29	LEMON LOVA	5.2758621
30	PURPLE HEART	7.2413793
31	LEMON	5.1379310
32	TWISTS	13.6206897
33	GREEN EYES	8.0689655
34	BLUEBERRY FRITTERS	8.8275862
35	VANILLA FROSTED	14.0344828
36	BLUE EYES	10.0000000
37	GLAZED DONUT	62.1724138
38	PLAIN CAKE	26.0344828
39	OLD FASHIONED	26.1724138
40	CHOCOLATE GLAZED	9.1724138

41	PLAIN OLD FASHIONED	7.6206897
42	BLUEBERRY CAKE	6.3793103
43	FRENCH CRULLERS	5.9310345
44	BEARCLAWS	5.2068966
45	CINNAMON ROLLS	4.8965517
46	BUTTERMILK BARS	4.7241379
47	YUM-YUM M&M	4.2758621
48	CINNAMON TWISTS	3.7931034
49	CHOCOLATE COCONUT	2.7241379
50	CHOCOLATE FROSTED CAKE	2.6896552
51	VANILLA CAKE	2.3103448
52	DOUBLE CHOCOLATE	1.5517241
53	OREO COOKIE	1.3448276
54	SALTY CARAMEL OLD FASHIONED	0.3793103
55	CHOCOLATE OLD FASHIONED	0.3448276
56	SID SPECIAL	0.0000000

A large orange circle is positioned on the left side of the slide, partially cut off by the edge.

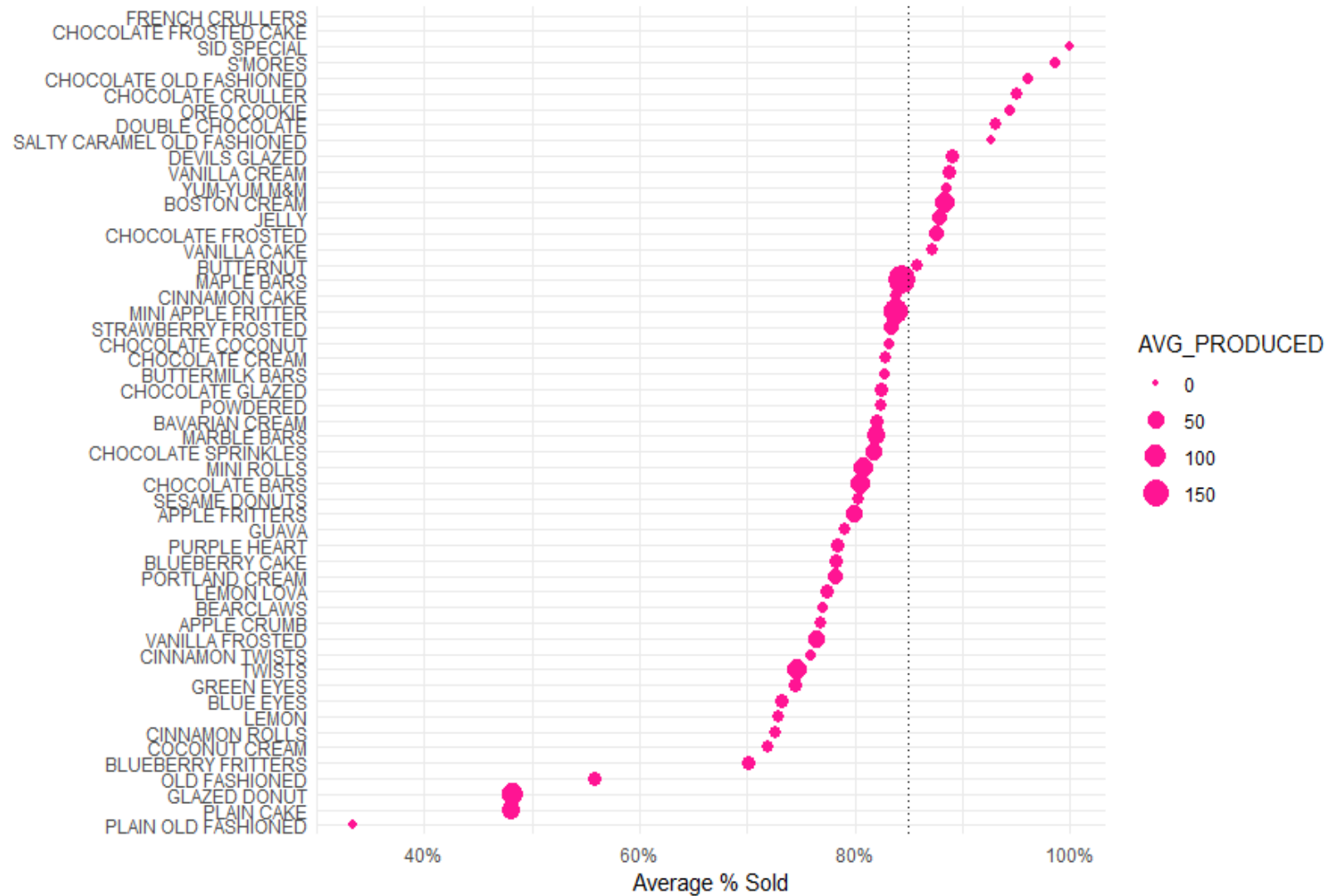
Updated Information

- Graphs and information with data collected past the initial 30 days of the observational study



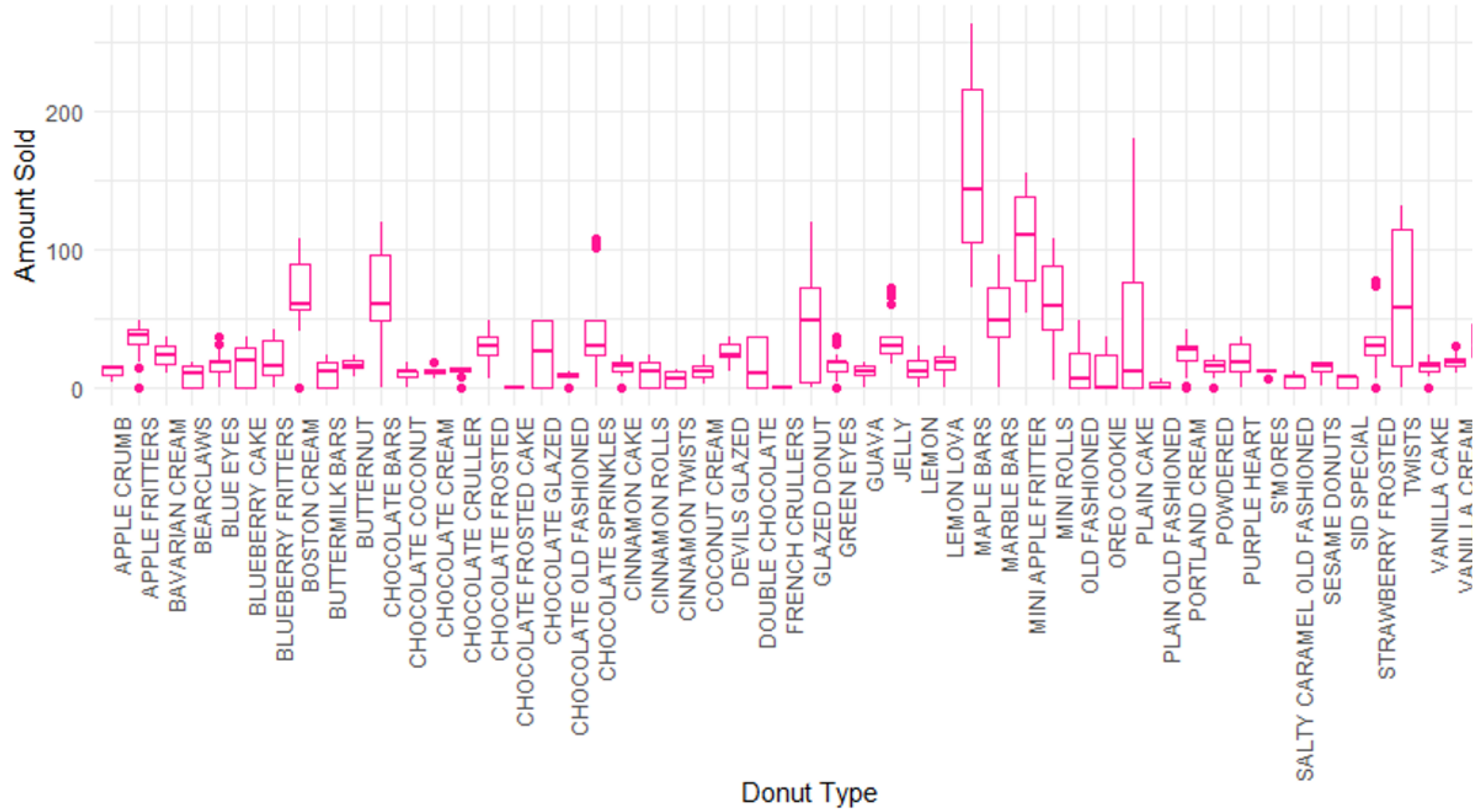
Average Produced in Accordance with Percent Sold

- The goal is to have 85% sold
- The extra 15% ensure the shop will not run out of a certain type
- Most Notably:
 - Plain cake : 48% sold from an average production of 60
 - Glazed : 48.2% sold from an average production of 92
 - Old Fashioned : 55.9% sold from an average production of 27
 - Twists : 74.6% sold from an average production of 73
 - Vanilla Frosted : 76.4% sold from an average production of 42



Average Amount Sold in Accordance with Donut Type

- The graph displays the mean donuts sold(produced-discarded) over the course of the study
- The best sellers of the donut types are as followed:
- Maple Bars, Plain Cake, Glazed, Mini Apple Fritters, Twists and Blueberry Fritters
- Among the best sellers, there was still significant variation shown by the large whiskers on the boxes



Data Quality Issues

- The values for the daily produced donuts were communicated poorly
 - There was unexplained overlap and independence in terms of some donut production
- In terms of using Discarded values or Production value
 - Discarded had more implementation error but more unique values
 - Production had less unique values thus higher possibility of error if those values were incorrect
- Issues led to some negative numbers when computing between the two
 - Solution: Given circumstances, changed negatives to 0 and kept change in mind when coming to conclusions